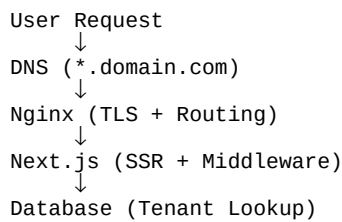


Dynamic Subdomain Multi-Tenant Platform

1. Overview

A scalable multi-tenant architecture where each user gets a unique subdomain mapped to shared infrastructure.

2. Architecture Diagram



3. DNS Setup

Use wildcard DNS to avoid per-user record creation.

```
*.domain.com → SERVER_IP
```

4. SSL Setup

Use wildcard SSL via DNS-01 challenge to avoid rate limits.

```
*.domain.com certificate
```

5. Nginx Configuration

```
server {
    listen 443 ssl;
    server_name *.domain.com;

    ssl_certificate /path/fullchain.pem;
    ssl_certificate_key /path/privkey.pem;

    location / {
        proxy_pass http://localhost:3000;
        proxy_set_header Host $host;
    }
}
```

6. Next.js Middleware

```
const host = request.headers.get("host");
const subdomain = host.split(".")[0];
```

7. Deployment Checklist

```
# 1. Setup DNS
Create wildcard record:
*.domain.com → SERVER_IP

# 2. Install Nginx
sudo apt install nginx

# 3. Install certbot + route53 plugin
sudo apt install certbot python3-certbot-dns-route53

# 4. Generate wildcard SSL
certbot certonly --dns-route53 -d *.domain.com -d domain.com

# 5. Configure Nginx
# (use config above)

# 6. Run Next.js
npm run build
npm start

# 7. Restart Nginx
sudo systemctl restart nginx
```

8. Conclusion

This architecture enables scalable, secure, and efficient multi-tenant systems using wildcard DNS, SSL, and application-layer routing.